神經外科系工作手冊

教學宗旨及目標

一. **宗旨**: 神經外科為外科內性質較為特殊之專科，神經外科針對輪訓至本科之醫師，除了強調各種神經外科常見疾病的臨床表徵與治療方式外，更強調面對重症病患與家屬的醫病關係與臨床問題的處理，責任心與榮譽感更為本科之基本要求。對於醫師輪調至神經外科，吾人認為教育之宗旨乃在於(1)傳授神經外科基本知識與技能(2)提供良好的臨床教學環境作為實、見習醫師進入臨床醫學的橋樑(3)建立以病患為中心之醫療服務(4) 針對神經外科從事有系統的回顧與複習(5)協助醫師自我認知，分析性向與生涯規劃。

二. 目標：所有至神經外科輪訓之醫師應該學習並充分瞭解之重點

1. 腦外傷與臚內出血之診斷與處理原則
2. 臚內高壓之治療
3. 脊椎外傷之診斷與處理原則
4. 退化性脊椎疾病之診斷
5. 腦血管疾病之診斷與處理原則
6. 對於臨床狀況之敏感度的提升
7. 處理臨床問題之邏輯思考
8. 學習與其他醫療團隊成員互相合作平等相處

神經外科工作規範

98-10-06

Orientation 時間:每月 1 日和 16 日，遇例假日則順延一天。

Orientation 地點：4:30PM 醫學大樓 2 樓神外 ICU-I 會議廳

一、 工作職責

每位 Resident & Intern 在病房及開刀房方面，跟隨指定主治醫師為主，
必要時，總醫師得以分配於開刀房與病房的學習處理其它主治醫師之病例。

A. 病房事項

1) Resident，Intern 及 Clerk 在早上 7:00 VS or CR 查房前，先自行訪視病人(負責病床區為所跟 VS 的所有床位)及收集 Data (如 I&O、CXR、CBC、Biochemistry)同時應熟悉病人現在情況及用藥.全病房 Invasive procedure (如 on Foley, on CVP, endotracheal intubation and chest tube insertion)，由 Resident 執行，或在 Resident 監督下由 Intern 執行。
2) Resident & Intern 負責病房時，除了一般的外科 history taking & P.E.外，在 review of system & P.E.要特別加入 Neuroexamination，including Coma scale, muscle power 等之檢查註記。
3) 每天早上查房時間須視各 VS 而有不同。
4) 白班 intern 及 resident 負責人接 7:30AM 至 4:30PM 之 new patient, 值班人員則負責處理 4:30PM 至 7:30AM 之 New patient,若有任何特殊病理或檢查要確實交接班。
5) 每天下班前(5:00PM)如有特殊檢查或情況病人，須與小夜 NSP & duty Resident or Intern 交班。
6) 與 NSP 配合，處理 order、Change dressing、拆線、拔除引流管等。並在 senior resident cover 指導下學習 invasive procedure,如 Lumbar puncture or CVP 等。
7) 每天紀錄一次 Progressive Note。出院病歷之完成, pre-op evaluation 及 post-op assessment 要確實記錄，
8) Pre-op order 需簡單病史回顧，診斷分析，影像檢查發現繪圖，註明手術方法及日期。
9) 若排為病房留守，則協助處理各項 order。
10) 準時參加開會（W3，W4 7:00AM 於 9H 外 R4 會議室）.

B. 開刀房事項

1) 白天處理所屬病人之 order 後，入開刀房學習無菌觀念、正確刷手、導尿、消毒、舖單、跟刀用心投入,積極發問。手術後幫忙傷口清潔及包紮,送病人至 POR。
2) 手術前一晚上, 對隔日開刀病人做評估,寫 Pre OP assessment ,由負責的 Resident or Intern 紀錄。
   - Pre OP assessment Form:
     a) Diagnosis
     b) Chief complaint
     c) Past history
     d) PE (Positive finding)，including GU system
     e) LAB data. EKG, echo, and imaging finding (drawing)

f) Op Indication
g) Plan

3) 手術房 11 房、12 房、13 房、15 房、18 房、19 房、20 房、21 房(醫學大樓 2F)
4) CR 得以視 OR 之需要,調配 Resident & Intern 至 OR 支援。

C. 值班事項

1) 值班時段由 4:30pm 至第二天的 7:30am,假日則為 8:00am 至隔天上班時間。
2) Duty doctor 負責接 duty time 的 new patient 及第一線處理各種 condition,若無法 handle,則由 CR cover。
3) Coma scale 檢查須確實, 若對處置有疑問,請充分了解病情後向專科 R 或 CR 報告.
4) B.P.高的病人須小心鑑別診斷是 Increased intracranial pressure (IICP) 所引致或確實是 Hypertension,勿隨意給降血壓的藥物。
5) EVD 不通時, 勿任意抽吸或注入, 請充分了解病情後向專科 R 或 CR 報告, 接受指示處理病人.
6) 第二天 duty 結束前,特殊病人之特殊 condition 須確實和當科 doctor 交班。

二、Activity:見每月學術活動表

常見之 Q&A
脊椎外傷:

(一) 脊椎外傷病人的入院必要處置

A. Immobilization:避免不必要的移動, 或使用輔助器具如固定頸椎, 長滑板等
B. Blood pressure: 病人易 hypotension, 所以要適量增加 I.V. fluid, 使用 Dopamin 維持血壓以 atropin 對抗心跳減慢等問題
C. O_2 或 intubation: 對頸椎外傷病人必須嘗試 on nasal Endo 以避免頸椎二度傷害
D. NG tube: 避免嘔吐與 aspiration pneumonia
E. Foley
F. Temperature control: 病人可能體溫過高或過低
G. Medication: 大量的 methylpredmisolone 有助於急性脊髓損傷病人復原,但請在 CR 允許下開立.

(二) 大量 Methylpredmisolone 給法

在脊椎外傷 8 小時以內給予
前 15 分鐘 bolus 給予 30mg/kg
停 45 分鐘後, 後 23 小時連續給予 5.4mg/kg/hr

(三) 何時須對脊椎外傷病人進行減壓手術(laminectomy)

對 incomplete spinal cord injury 的病人, 有下列狀況可考慮開刀
a. 神經功能惡化中
b. 在 CT, Myelo or MRI 檢查看到 canal 內有 bone chip, HIVD, Hematoma 等 mass lesion
c. 有 fracture-dislocation,例如:locked facets 等
d. 病人有明顯的 radiculopathy (單一 root 被壓迫).

(四) 一般脊椎手術有哪些種類

脊椎手術可分為下列幾大類:
(A) 減壓: 對 spinal cord 減壓 (cervical or lumbar laminectomy)
對 root 減壓 (open disectomy, microdisectomy)
(B) 固定:對脊椎滑脫、骨折的病人內固定
前固定: Cervical – Odontoid screw, Caspar plate
Thoracic & Lumbar -- Kaneda
後固定: Cervical-- Lateral mass plate, transpedicle screws, tranarticular screws
Thoracic—Hook screw
Lumbar—Transpedicle screws
(C) 骨融合: Cervical or lumbar cage
(D) 人工椎間盤 Bryan disc, Prestige, Pro Disc C

(五) 是否脊椎手術皆須備血

本科進行 Cervical and lumbar microdisectomy 與 cervical laminectomy 流血極少，不須備血，但其餘脊椎手術仍須備血.

(六) 脊椎手術病人術後要注意什麼?

脊椎手術病人手術後有可能因傷口內血腫壓迫到 spinal cord 而使病人四肢癱瘓，所以手術後一定要注意病人是否有四肢無力、麻木與傷口疼痛的情形. 其餘的問題如下:

a. urine and stool: 病人可能大小便困難
b. Heart rate increase, urine decrease, BP drop & Hb drop: 病人可能 blood loss 過多
c. 呼吸: 頸椎手術後病人容易影響呼吸
d. pain: 可以問病人是否願意自費裝置 PCA (bolus demerol 會影響呼吸、嘔吐、過敏等副作用)

(七) 脊椎病人手術後，有哪些輔助外固定器

頸椎外固定器:
a. Halo-jacket： 固定效果最好， 但安裝不容易
b. Four poster
c. Miami neck collar, Hard neck collar
d. Soft neck collar
腰椎外固定器: body jacket, Taylor back brace

(八) 對脊椎病人有什麼影像學檢查

a. X-ray: AP, Lat, Flexion-extension (dynamic view), Oblique
b. Myelogram: 須打顯影劑於 spinal canal 內，屬侵入性檢查
c. CT:對 bone 看的比較清楚，若與 myelogram 一起做，效果同等 MRI
d. MRI: 可以對 soft tissue 有很清楚的分辨效果.

脊椎外傷

(A) Neurological Assessment :

1. Completeness of lesion
   (a) Definition of incomplete lesion: any residual motor or sensory function below the level of injury, e.g.: sacral sparing; voluntary movement in Les.

   (a) Three basic pattern of incomplete lesions:
   Brown-Sequard syndrome
   Central cord syndrome
   Anterior cord syndrome

2. Level:

3. Spinal Shock
   (a) Transient loss of all spinal reflexes below the level of the injury → flaccid paralysis. It may last for varying periods (2 weeks to several months), followed by spasticity below the level of the lesion.

(b) Another condition that also call " spinal shock" – hypotension (shock) that follows spinal cord injury. It may be caused by skeletal muscle paralysis and loss of vascular tone ± blood loss

4. Motor level assessment:

Segment	Muslces	Action to test
C5	Deltoid or biceps	Shoulder abduction nor elbow flexion
C6	Wrist extensors	Cock up wrist
C7	Triceps	Elbow extension
C8	Flexor digitorum prof	Squeeze hand
T1	Hand intrinsics	Abduct little finger
LL32	Hiopsoas	Flex hip
L4	Quadriceps	Straighten knee
L5	Tibitalis anterior	Dorsiflex foot
S1	EHL	Dorsiflex big toe
	Gastrocnemius	Plantarflex foot

C4: Diaphragm, T2-9: Intercostal muscle
Muscle strength grading

Grade	Strength
0	No contraction
1	Flicker or trace contraction
2	Active movement with gravity eleiminated
3	Active movement against gravity
4	Active movement against resistance 4-: slight resist 4: moderate resist 4+: strong resist
5	Normal strength

(5) Sensory level assessment
Key sensory landmarks

Level	Dermatoma
C4	Shoulder
C6	Thumb
C7	Middle finger
C8	Little finger
T4	Nipple
T6	Xiphoid
T10	Umbilicus
L3	Just above patella
L4	Medial malleolus
L5	Great toe
S1	Lateral malleolus
S4-5	Perianal

* Sensory segment "jump" from C4 to T2 on the upper chest-neck junction area

(6) Reflex test:

Segment	Reflex

C5	Biceps
C6	Brachiradialis
C7	Triceps
T9,10	Abd. cutaneous reflex
T11,T12	Abd. cutaneous reflex
L2,3	Cremasteric reflex
L3,4	Knee jerk
S1	Ankle jerk
S2,3,4	Bulbocavernous reflex Priapisum Anal cutaneous reflex

(B) Management of Spinal Injury

Spinal (cord) injury should be considered and treated in any suspected victims until proven otherwise. e.g.: trauma, unconscious patient.

(1) Management in the field:

1. Immobilization prior to moving the patient
2. Maintain blood pressure
3. Maintain Oxygenation
4. Brief motor examination (arms, hands, legs, toes)

(2) Management in the hospital

1. Immobilization (Neck collar, back board)
2. Hypotension treatment:
   (a) Intravascular volume maintain (isotonic fluid)
   (b) Pressor, if necessary (Dopamine)
   (c) Atropine for bradycardia with hypotension
3. O2 or careful intubation : much attention for respiratory function in upper cervical injury
4. NG decompression
5. Foley cather to prevent bladder distension
6. Temperature regulation
7. More detailed neuro-evaluation
8. Radiographic evaluation
9. Medical management specific to spinal cord injury. e.g.: High dose methylprednisolone

(3) Detailed Neuro-evaluation included

(A) Focused history:

1. Mechanism of injury (hyperflexion, extension, axial loading ---)
2. LOC (?)
3. Hisotyr of transient weakness or numbness following trauma

(B) Palpation of the spine for tenderness or "step-off"

(C) Motor level assessment:

1. Skeletal muscle examination
2. Rectal examination

(D) Sensory level assessment

1. Sensation to pinprick (spinothalamic tract)

2. Light touch (anterior cord function)
3. Propioception /joint position sense (Posterior column)

(E) Evaluation of reflex
1. Muscle stretch reflex
2. Abdominal cutaneous reflex
3. Cremasteric reflex
4. Bulsocavernous reflex, anal cutaneous reflex

(F) Examine for signs of autonomic dysfunction
1. Altered patens of perspiration
2. Bowel or bladder incontinence
3. Priapism

(4) Methylprednisolone (Solumendrol)
1. Beneficial when given within 8 hours after injury
2. Bolus: 30 mg/kg over 15 min
3. Maintenance: 5.4 mg/kg/hr

(5) Radiographic evaluation
1. C-spine (AP +Lat): must include C7-T1 junction (Swimmer' s view, if necessary)
2. T-L spine
3. Dynamic study (flexion – extension view)
   -- to disclose occult instability
   -- contradication:
   (a) Not cooperative patient
   (b) Any subluxation > 3.5 mm
   (c) With significant neurlogic deficits
4. CT scan (with or without myelograph): useful in upper C-spine injury, Horizontal fracture, obese patient.
5. Emergent MRI: indication:
   (a) Incomplete lesion with normal alignment
   (b) Neurologically deterioration
   (c) Fracture level incompatible with deficit level
   (d) R/O soft tissue compression. E.g.: HIVD, Hematoma

(6) Cranial-Cervical traction

1. Purpose: reduce fracture-dislocation
   (a) Maintain normal alignment
   (b) Immobilize the spine
   (c) Decompress the cord
2. Common : Gardner-Well tong
3. Pin site: 2-3 finger breadths above pinna (directly above EAM)
4. Initial weight (Ibs) = 3 x cervical vertebral level
   (a) 5-10 lb increment until desired alignment
   (b) Normal: < 10 Ibs per vertebral level
   (c) 1 Ib= 0.454 kg
5. Be care of Over-distraction

(B) Upper C-Spine Injury (C1-C2)

(1) Atlanto-occipital dislocation: Usually minimal neurological deficit or cervical-medullary

dislocation (respiratory arrest).

(2) Jefferson fracture

1. C1 arch fracture
2. Usually from axial loading
3. Usually no neurologic deficit
4. Evaluation : Open-mouth view CT
5. Usually no operation needed, but need brace immobilization

(3) C2 odontoid fracture

Type	Characteristics	Stability	Treatment
I	Tip fracture (above transverse ligament)	Unstable	Usually need Op.
II	Neck base fracture	Usually unstable	Controversial
II-A	Neck base fracture with large bone clips	Usually unstable	Early Operation is recommended
III	Through C2 body	Usually unstable	90% heal after extremity immobilization

Indication for type II odontoid fracture

Age $\geq$ 7 Y/O and any of the following:

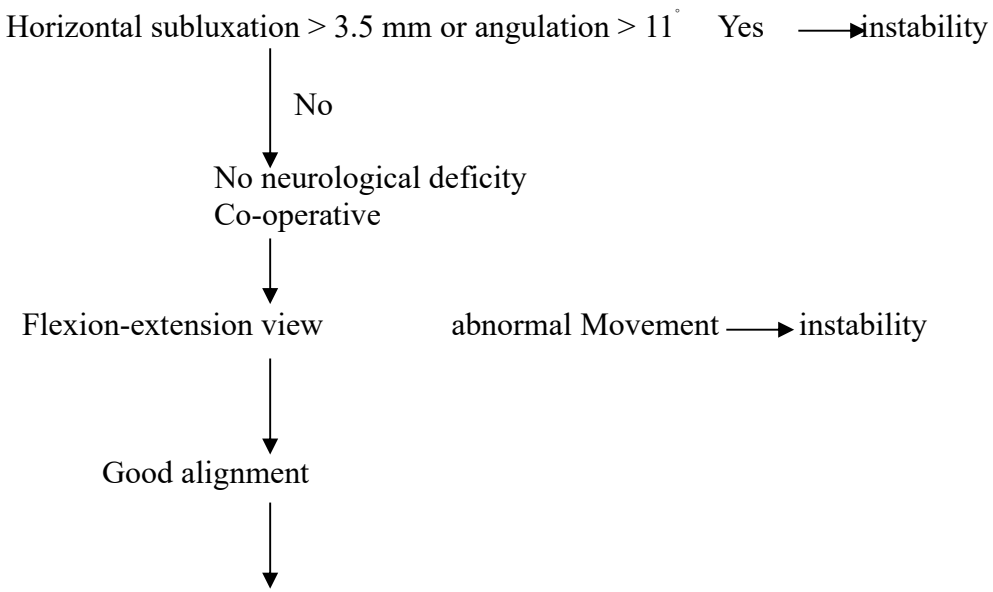
(a) Displacement $\geq$ 6 mm
(b) Instability at the fracture site in the Halo vest
(c) Non-union

(4) Hangman' s fracture (traumatic spondylolithesis of the axis)

1. Bilateral C2 pedicles fracture
2. Often with C2-C3 subluxation
3. usually stable
4. 90% heal with immoblization

C3-C7 Subluxation and Fracture

1. Stability



Followed after muscle spasm

(2) Tear-drop fracture

1. With neurologicaldeficit
2. Bone fragment is large
3. Vertebral body malalignment
4. Bone fracture on sagittal plane
5. Significant soft tissue swellay

} all suggest instability

If no above problems ⟶ flexion-extnesion view

(3) Facet block

1. Unilater or bilateral
2. Try closed reduction first (cranial-cervical taction)
3. Open reduction
4. Internal fixation or external fixation

(4) Whiplash syndrome

1. Soft tissue injury without bony lesion
2. Occult instability must be ruled out
3. paraspinal muscle spasm is significant
4. Philadelphia collar for protection repeat films if pain persists 1-2 weeks later

(5) SCIWORA (Spinal Cord Injury Without Radiographic Abnormality)

1. Usually in child
2. Admit patient to hospital
3. Bed rest with rigid neck collar
4. MRI to document presence of spinal cord injury
5. Detailed discussion with patient & family about seriousness of injury
6. Immobilization in brace at least 3 months
7. Prohibition of contact & con-contact sports
8. Regular follow-up
9. Repeat flexion-extension study 3 months later

(6) Cervical Braces

1. Collars; soft collar, Philadelphia collar
2. Poster brace: 4-poster brace (good for prevention flexionat mid-cervical levels)
3. Cervico-thoracic or thosees (COT)
4. SOMI brace (Sternal occipital Mandibular Immoblization)
5. Halo-vest brace

Cervical Disc Herniation

(A) Symptoms

1. Discogenic pain syndrome
2. Radiculopathy
3. Myelopathy
4. Radiculo-myelopathy

(B) Radiculopathy

1. Provocative test: Lhermitte' s sign, Spurlng' s sign, Shoulder abduction test
2.

	C4-5	C5-6	C6-7	Ct-T1
Root compressed	C5	C6	C7	C8
Reflex	Deltoid	Biceps brachirodialis	Triceps	Finger jerk
Motor weakness	Deltoid	Forearm flexion	Forearm extension	Hand intrinsic
Paresthesia hypesthesia	Shoulder	Upper arm thumoradical forearm	Finger 2 & 3	Fignger 4 & 5

(C) Myelopathy

1. Incomplete cord syndrome: Central cord syndrome, Brown-Sequard syndrome, anterior cord syndrome
2. Long tract sign (Babinski sign, Hoffman sign)
3. Spascity of lower extremities (Spastic paraparesis)
4. Sphincter function impairment

(D) Radiculo-myelopathy

Radiculopathy + Myelopathy

1. Pattern of disease course
   (a) 75% Step-wise deterioration
   (b) 20% Stationary
   (c) 5% Fluctuation

2. Evaluation
   (a) C-spine 4 views + flexion-extension view
   (b) Myelography + CT
   (c) MRI

3. Prognosis
   (a) Duration of deficit (most important)
   (b) Severity
   (c) Age

4. Treatment
   (a) Neck collar immobilziation

(b) ACDF (Anterior Cervical Discectomy with Fusion)
(c) Posterior laminectomy

5. Post-op. check
   (a) Evidence of post-op. hematoma
      1. Respiratory distress
      2. Extreme swallowing difficulty
      3. Tracheal deuriation
      4. Wound swelling
   (b)Weakness of nerve root of level operated
   (c) Long tract signs
   (d) Position of bone fusion graft
   (e) Hoarseness (recurrence laryngeal nerve injury)

Degenerative spine disease (DSD)

DSD is a progressive deterioration of the structures of the spine including:

I. disc abnormalities:
   A. the proteoglycan conent of the disc nucleus decreases with age
   B. disc loss of hydration occurs
   C. tears develop in the disc annulus progress to internal disruption of the lamellar architecture. Herniation of the nucleus may occur
   D. mucoid degeneration and ingrowth of fibrous tissue ensues
   E. subsequently disc resorption occurs
   F. there is a loss of disc space height and increased susceptibility to injury
II. facet joint abnormalities: hypertrophy and capsular laxity
III. osteophytes often form on the edges of the VB bordering the degenerated disc
IV. spondylolisthesis: subluxation of one VB on another
V. spondylolysis: alternative term for isthmic spondylolisthesis, a failure of the neural arch due to defect in the pars interarticularis which may present as spondylolisthesis
VI. hypertrophy of the ligamentum flavum

Clinical presentation:

1. the above abnormalities may produce spinal stenosis which can lead to neural compromise producing the following symptoms:
2. radicular symptoms
3. neurogenic claudication(lumbar) or spinal myelopathy(cervical

etiology: multifactorial

1. culmulative effects if microtrauma and macrotrauma to the spine
2. osteoporosis
3. in the lumbar spine: stress on the spine including the excess body weight; loss of muscle tone (primarily abdominal and paraspinals) resulting in increased dependence on the body spine for structural support
4. cigarette smoking: epidemiologic studies showed incidence is higher

spondylolisthesis
anterior subluxation of one vertebral body on another, usually L5 on S1, occasionally L4 on L5.

Grade I:< 25%; Grade II:25-50%; Grade III: 50-75%; Grade IV:>75%
CLASSIFICATION OF SPONDYLILISTHESIS

1. isthmic spondylolisthesis: a failure of the neural arch due to defect in the pars interarticularis ("Scotty dog" on the oblique L-S x-ray)
2. dysplastic: congenital. Upper sacrum or arch of L5 permits spondylolisthesis. No pars defect
3. degenerative: due to long-standing intersegmental instability. No break in the pars.
4. Traumatic: due to fractures in other areas of the body hook and the pars
5. Pathologic: generalized or local bone disease

Recommended treatment for spondylolisthesis
1. Degenerative: decompression +/- fusion
2. Traumatic: decompression plus fusion

Spinal stenosis

1. Central canal stenosis: narrowing of the AP dimension of the spinal canal. The reduction in canal size may cause local neural compression and/or compromise of the blood supply to the spinal cord (cervical) or the cauda epina (lumbar)
2. Foraminal stenosis: narrowing of the neural doramen
3. Lateral recess stenosis (lumbar spine only)

Lumbar spinal stenosis

Key points

1. caused by hypertrophy of facets and ligementum flavum, may be exacerbated by disc bulging or spondylolisthesis, may be superimposed on congenital narrowing
2. most common at L4-5 and the L3-4
3. symptomatic stenosis produces gradually progressive back and leg pain with standing and walking that is relieved by sitting or lying (neurogenic clauducation)
4. symptoms differentiated from vascular claudication which is usually relieved at rest regardless of position
5. usually responds to decompressive surgery; fusion may be an adjuant

Symptomatic lumbar stenosis is most common at L4-5, the L3-4, L2-3 and lastly L5-S1. It is rare at L1-2.
Stable form: hypertrophy of facets and ligementum flavum accompanied by disc degeneration and collapse
Unstable form: have above with superimposed of degenerative spondylolisthesis or degenerative scoliosis
The neurologic exam is normal in about 18% if cases (including normal muscle stretch reflexes and negative straight leg raising). Absent or reduced ankle jerks and diminished knee jerk is common. Pain may be reproduced by lumbar extension.

Radiographic evaluation: lumbosacral spine x-rays; CT scan; Myelogram; MRI.
Treatment :
Nonsurgical management: NSAID and physical therapy
Surgical decompression is undertaken when symptoms become severe in spite of medical management. The goals of surgery are pain relief, halting progression of symptoms, and possibly reversal of some existing neurologic deficit. Posterolateral fusion is recommended for patients with stenosis and associated degenerative spondylolisthesis who required decompression.

Cervical spinal stenosis

Three types of clinical problems:

1. nerve root compression may cause radicular complaints
2. spinal cord compression may cause myelopathy
3. pain and paresthesias in the head, neck and shoulders with little or no suggestion of radiculopathy nor abnormal physical findings. The group is the most difficult to treat, and often requires a good physician-patient relationship to decide if surgical treatment should be undertaken in an attempt to provide relief.

Syndromes:

1. transverse lesion syndrome: involvement of corticospinal and spinothalamic tracts and posterior columns, with anterior horn cells segmentally involved.
2. Motor system syndrome: primarily corticospinal tract and anterior horn involvement with minimal or no sensory deficit.
3. Central cord syndrome: motor and sensory deficit affecting the UEs more than the Les. This syndrome is characterized by dysfunction of the watershed areas located centrally within the cord, which may be responsible for prominence of hand syndromes. Lhermitte's sign may be more common in this group.
4. Brown-equard syndrome: often with asymmetric narrowing of the canal with the side of greater narrowing producing ipsilateral corticospinal tract and posterior column dysfunction with contralateral loss of pain and temperature sensation.
5. Brachialgia and cord syndrome: radicular UE pain with lower motor neuron weakness, and some associated long tract involvement (motor and/or sensory).

Treatment:

Nonoperative management: consists primarily of cervical bracing in an attempt to reduce motion and hence the cumulative effects of trauma on the spinal cord.

Surgical treatment:

Indications:

A patient presenting with myelopathy that is already established, and has been present for an unknown length of time without suggestion of progression may not derive any benefit from surgery. Indications for surgery are primarily patients with radiologic evidence of spondylotic degeneration of the cervical spine in the following settings:

1. patients with progressive myelopathy who may be neurologically stabilized with surgery
2. patients with severe pain often have pain relief with surgical decompression
3. patients presenting early in their course fare better than whose with advanced and longstanding disease.
   Anterior or posterior approach is depends on patient's condition.'
   Coincident symptomatic lumbar and cervical spinal stenosis is usually managed by first decompressing the cervical region, and later operating on the lumbar region (unless severe neurologic claudication dominates the picture). It is also possible, in selected cases, to operate on both in a single sitting.

Pre-op order
Related image study
NPO for 8 hours, set iv line, CXR, EKG, Sign operation and anesthesia permit
Check CBC/DC, PT/APTT, BUN, Cr, Na, K, AST, ALT, sugar
Prepare blood : pRBC and FFP in most cases except microdisectomy, anterior cercial disectomy and vertebroplasty
Bring medication to OR: cefazolin 2pc, dexan 2pc, surgical 1 pc **(** a hemostatic agent

(blood-clot-inducing material) made of an oxydized cellulose polymer)

昏迷:

昏迷指數之評估: 乃評估意識狀態之程度, 而不是判斷局部神經功能缺損

(一) GCS (Glasgow Coma Scale)(適用於 4 歲以上)

	最佳睜眼反應	最佳語言能力	最佳活動反應
6			可依令而行
5		人、時、地、物都清楚	可碰觸痛刺激
4	自行睜眼	混亂	逃離痛刺激
3	聽到聲音	可理解之字句	屈曲反應
2	痛刺激	呻吟	伸張反應
1	無反應	無反應	無反應

(二) Children`s Coma Scale(適用於 4 歲以下)

	Best Eye	Nonverbal child	Verbal child	Best Motor
6				Obey
5		smiles, oriented to sound, follows objects, interacts	oriented and converses	Localize pain
4	Spontaneous	consolable when crying and interacts inappropriately	disoriented and converses	Withdraws to pain
3	To speech	inconsistently consolable and moans; makes vocal sounds	inappropriate words	Flexion
2	To pain	inconsolable, irritable and restless; cries	incomprehensible sounds	Extension
1	None	None	None	None

(三) 解剖學上造成昏迷的原因可能有

1. Higher brain stem (central upper pons or mid brain)之功能異常
2. Bilateral diencephalons 之功能異常
3. Diffuse lesion in both cerebral hemisphere (cortical or subcortical white matter)

(四) Coma 之原因

1. Toxic/Metabolic causes of Coma
   (A) Electrolyte imbalance: Hypo-hypernatremia, hypercalcemia, ↑ BUN
   (B) Endocrinopathy:hypoglycema, DKA, HHNK, Adrenal crisis, myxedema
   (C) Vascular: vasculitis, DIC, Hypertensive encephalopathy
   (D) Toxic: CO, drug (nacrotic ---) Pb
   (E) Infection /Inflammatory: meningitis, encephalitis, sepsis
   (F) Neoplastic: leptomeningeal carcinomatosis
   (G) Nutritional: Wernicke`s encephalopathy, Vit B12 def.
   (H) Inherited metabolic disorder: porphyria

(I) Organ failure: uremia, hepatic coma, hypoxemia
(J) Epileptic: status epileptic

(2) Structure causes of coma
(K) Vascualr
(i) bilateral corticalor subcortical infarct
(ii) occlusion of vessel supplying both cerebral hemisphere
(iii) bilateral diencephalic infarct
(iv) infectious
(L) Neoplastic
(M) Trauma: contusion, edema
(N) Herniation from mass effect
(O) IICP
(P) Acute lateral shift of brain

(3) Pseudo-coma (Psychiatric, lock-in, neuromuscular weakness)

昏迷病人之檢查與治療

1. ABC (Airway, Breathing, Circulation)
2. 抽血檢查 CBC (D/C), Sugar, BUN, Cr, Na, K--, ABG, Drug(ex. Dilantin)
3. Emergent supportive medicine: glucose, thiamin, naloxone
4. Core neurological examination: pupil, EOM, eye reflex, motor response, respiration
5. 是否有 herniation syndrome
6. 是否有 meningeal sign(Nuchal rigidity, fever ---)
7. Treat generalized seizure
8. Treat metabolic abnormalities (acid-base, electrolyte, body temp)
9. 詳細病史之探查(patient`s family, witness)
10. 針對病因做特定之治療

Herniation Syndromes:

1. Central herniation
2. Uncal herniation , supra-tentorial herniation
3. Subfalcine hernation
4. Upward herniation, infratentorial herniation
5. Tonsillar herniation

顱內壓上升(IICP)

1. 正常值:一般大人和大小孩: ICP≤10-15 mmHg, 頭部外傷造成的 IICP 一般認為≥16 mmHg 就須給予治療
2. 症狀; 頭痛、嘔吐、意識不輕、視乳突水腫. Cushing triad: Hypertension , Bradycardia, Irregular respiration, full triad 僅約１／３patient 有
3. 原因:腦水腫、腦瘤或腦血塊、水腦症、腦血流過多(hyperemia)、顱靜脈竇栓塞；不明原因

4. 診斷:
   (1) ICP 監視器
   (2) Brain CT: third ventricle, perimesencephalic cistern, basal cistern 變窄或消失, midline shift
   (3) Lumbar puncture:必須在 brain CT 做完，謹慎評估下才可執行
5. 治療 :
   (1) Hyperventilation: keep PaCO2 25-30 mmHg
   (2) Hyperosmolar therapy: (Mannitol, Glycerol, Lasix ---)
   (3) Head elevation : 30° – 45°
   (4) CSF drainage
   (5) Adequate sedation, control HTN
   (6) Barbiturate coma
   (7) Hypothermia
   (8) Surgical removal of mass, if possible
   (9) Steroid

頭部外傷

Classification of head injury

(1) By mechanism
  (a) closed head injury
  (b) Penetrating head injury (Gun shot, other open injuries)
(2) By severity
  (a) Mild head injury: GCS 13-15
  (b) Moderate head injury: GCS 9-12
  (c) Severe head injury: GCS 3-8
(3) By morphology
  (a) Skull fracture
    Vault fracture (linear or depressed)
    Basilar fracture (CSF leakage, cranial nerve palsy)
  (b) Intracranial lesions
    Focal (epidural , subdural, intracerebral)
    Diffuse (concussion, coup & counter-coup contusion, diffuse oxonal injury)

(一) Mild Closed Head Injury

Definition: The patient is awake, and may be orientated (GCS 13-15)

Initial Work-up:

History
(1) Name, age, sex, race, occupation
(2) Mechanism of injury
(3) Time of injury
(4) LOC immediately post-injury
(5) Subsequent level of alertness
(6) Amnesia: retrograde, antegrade
(7) Headache: mild, moderate, severe; vomiting
(8) Seizure
General examination
Limitted neurological examination
Skull X-ray
C-spine & other radiography as indicated
Blood alcohol level and urine toxic screen
CT scan of the head should ideally be ottained in best but the completely asymptomatic p’ t

After Initial Work-up

Admission, if there is—
1. Significant amnesia
2. History of LOC, seizure
3. Deterioration of consciousness
4. Moderate to severe headache
5. Significant alcoholic/drug intoxication

6. Skull fracture, sign of basilar skull fracture
7. CSF leakage
8. Significant associated injuries
9. No reliable companion at home
10. Abnormal brain CT

Discharge, if ---
1. The patient does not meet any of the criteria for admission
2. Discuss need to return if any problems develop and issue a warning sheet (head injury sheet)
3. Schedule follow-up clinic visit, usually within one week

(二) Specific Examination for Head Injury Patient

(A) General PE
1. Granium inspection & palpation
(a) Evidence of basilar skull fracture
Raccoon`s eyes, Battle`s sign, CSF rhinorrhea/otorhea hemotympanum
(b) Check for facial fracture
Le Fort fracture
Orbital rim fracture
2. Cranio-cervical auscultation
(a) Carotid artery: bruit of carotid dissection
(b) Around the eye: bruit for CCF
3. Physical signs of spinal trauma

(B) Neurologic Examination

1. Cranial nerve examination:
(a) Pupil; resting position, size, light response
(b) Cranial nerve palsy: oculomotor palsy, abducent palsy, facial palsy ---
(c) Fundus examination

2. Level of consciousness/ mental staus
(a) GCS
(b) Orientation
3. Motor examination
(a) All 4 limbs motor strength, cooperative patient
(b) Motor response to noxious stimulus
(c) Rectal examination, anal tone, bulbocavernous reflex –

4. Reflexes

(a) DTR
(b) Plantar reflex (Babinski sign)
(c) Anal wink & bulbocavernous reflex

Admitting Orders for Mild Closed Head Injury

1. Activity: bed rest with head elevation 30˚

2. Vital sign, GCS q.2.h
3. NPO until alert; then clear liquids, advance as tolerated
4. Isotonic IVF urn at maintenance (average size adult:=75 cc/hr, ped: 1000-1200 cc/m^2/day)
5. Mild analgesics: acetaminophen; codeine p.r.n.
6. Anti-emetics, carefully used if indicated (avoid over sedation)
7. Notify duty Dr. if conscious change

(三) Scalp Injury & Subgaleal Hematoma

Five layers of scalp in cross section "SCALP"
1. **S**kin
2. sub**C**utaneous fascia
3. Galea **A**poneurosis
4. **L**oose areolar tissue
5. **P**ericranium

Vascular anatomy of scalp
5 major arteries supply the scalp on each side, which anastomosed each other freely & profoundly.
Mainly located in scalp Just above galea
1. Supra-orbital artery
2. Supra-trochlear
3. Superficial temporal artery (largest)
4. Posterior auricular artery
5. occipital artery

Scalp injury
1. Simple linear
2. Large, stellate with or without tissue defect – may need flap reconstruction (sonsultation with Plasty Dr.)

Management of simplest linear scalp laceration
1. Focal compression strongly before closure (to avoid massive bleeding)
2. Shaving around the lesion
3. Extensive debridement & irrigation
4. Two layer suture
Galea – absorbable suture is better
Skin – monofilament Nylong
5. Tetanus toxoid
6. Drains, rarely necessary
7. Antibiotics, may be not necessary if wound is clear

Subgaleal hematoma

2. May cross sutures (D/D with cephalhematoma)
3. May occur without bony trauma
4. Soft, fluctuate mass. Usually not communicated with CSF

Treatment

1. Almost never require treatment beyond analgesics
2. Avoid percutaneously aspiration
3. Closely watch for intra-vascular content, serial hemoglobin & hematocrit check is necessary in large lesions
4. If persist > 6 weeks, obtain a skull film

(四) Moderate Head Injury

Definition : GCS 9-12

Initial work-up:
1. Same as for mild head injury, except that baseline blood work (CBC, Chemo, PT/APTT) may be obtained
2. CT scan is advisable in all cases
3. Admission for observation is the safest option, even if the CT scan is normal

Admission order
1. Orders as for minor head injury. (NPO, if surgicalintervention is intended)
2. More frequent neurologic checks (q.1.h, better)
3. Admit to NSICU(I) or NSICU(II), if Ct shows any significant abnormality
4. Follow-up CT scan if condition deteriorates or preferably prior to discharge

Hyper-osmolar therapy for IICP

Mannitol
(A) Contra-indication:
1. Hypotnesion
2. Relative: allergy, ---

(B) Indication:
2. Evidence of herniation (localizing sign)
3. Evidence of mass effect (focal deficits)
4. Sudden deterioration prior to CT
5. After CT: if a lesion responsible for IICP is identified any on going to OR
6. To assess “salvageability”

(C) Dosage:
1. Bolus: 1gm/kg, over 20 min
2. Onset of effect: 1-5 min, peak:20-60 min
3. Maintance: 0.25 – 0.5 gm/kg, q 4-6 hr

(伍) Severe Head Injury

Definition : GCS 3-7

Management;
(A) History

1. Age of patient and type and time of accident
2. Drug or alcohol intake
3. Neurologic progression
4. Vomiting, aspiration, anoxia, seizure
5. Past history: including medication & allergies, CRF(?), anticoagulant ---

(B) Cardiopulmonary stabilization
1. Airway: intubate early
2. Blood pressure: normalize promptly using saline or blood
3. Catheters: Foley, NG (or OG), CVP
4. Diagnostic film: Brain CT, Chest X-ray, C-spine, Abdomen, Pelvis, Extremities, if indicated

(C) General examination

(D) emergency measures for associated injuries
1. Tracheostomy
2. Chest tubes
3. Spine stabilization: collar, traction
4. Abdominal paracentesis

(E) Neurological examination
1. GCS (eye opening, motor response, verbal response)
2. Pupil (resting position, movement, size, light reflex)
3. Oculocephalic response (Doll' s eye)
4. Oculovestibular response (Caloric test)

(F) Therapeutic measure
1. Surgical intervention for mass lesion, responsible
2. Hyperventilation, hypersomolar tyerapy, steroid (?), CSF drainage
3. Adequate sedation
4. ICP & blood flow monitor
5. Dilantin (?)
6. Barbiturate therapy
7. Hypothermia

Therapy for Severe Head Injured Patient

(A) General supportive measures
1. Adequate CPP (MAP-ICP), avoid hypotension, hypoxemia
2. Seizure prophylaxia
3. Sedation & paralysis (codeine, propofol ---)

(B) Treatment of IICP
1. Hyperventilation
   PaCO2: 25-30
   Onset of action : $\leq$30 second, peak effect = 8 min
   Over hyperventilation should be avoided
   Prolonged hyperventilation (> 72-96 hrs) is not advocated
2. CSF drainage
   Posture drainage (head elevation 30º - 45º)

Surgical drainage (ventricular puncture ---)
3. Hyperosmolar therapy with/without diuretics
4. Steroid (value in head injury is not clear)
5. Barbiturate therapy

(C) Systemic & medical care

1. Pulmonary care (hypoxia, pneumonia, neurogenic, pulmonary edema ---)
2. Cardiovascular care
3. Deep vein thrombosis & pulmonary emboli
4. Electrolyte abnormality (especially hyponatremia)
5. Coagulapathy & DIC
6. UTI & bed sore prevention
7. Nutrition

Specific Types of Head Injury

(A) Depressed skull fracture

1. Better seen in brain CT (bone window)
2. Criteria to elevate depressed fracture
   (a) > 8-10 mm depression (or > thicknes of skull)
   (b) Deficit related to underlying brain
   (c) CSF leakage or pneumocranium (dura defect)
   (d) Compound depressed fracture

(B) Epidural (Extradural)Hematoma

1. Typical presentation
   (a) Brief ILOC → lucid interval → progressive deterioration (contralaternal hemiparesis + ipsilateral pupil dilation)
   (b) Most often site: temporal area (middle meningeal artery tear) other sites are usually possible
   (c) Sources: 85% arterial bleeding (dura vessels)
   Others: bone marrow bleeding, venous sinus rupture (eg. Bifrontal EDH due to SSS rupture of supra & infra-temporial EDH due to transverse sinus rupture)
   (d) Classical CT presentation: biconvex high density lesion beneath the skull, homogenous or heterogeneous density
   (e) Optimal early diagnosis & prompt surgical evacuation can decrease the morbility & mortalty maximallly.

** An intracranial mass that produce more than 5 mm of midline shift or is large in volume (over 30 ml in supra-tentorial space or 10 ml in infratentorial fossa) has a much greater risk of developing danagerous brain herniation .

(C) Subdural Hematoma (acute and subacute)

1. Source of bleeding
   (a) Avulsed bridging vein

(b) Ruptured cortical vessels
(c) Lacerated brain

2. More impact force is needed to produce acute SDH than EDH, that makes SDH more lethal
3. Time frame & density on CT for SDH:
   (a) Acute SDH: 1-3 days, hyperdensity
   (b) Subacute SDH: 4 days – 2~3 weeks, isodensity
   (c) Chronic SDH: > 3 weeks, iso to hypodensity
4. Shape of acute SDH in brain CT:
   More diffuse, less uniform, usually concave over brain surface
5. Treatment:
   (b) Rapid surgical evacuation for symptomatic SDH greater than 1 cm at the thickest point is considered
   (c) Consider conservative treatment if smaller hematoma, severe hemispheric swelling that surgery may increase the brain injury, or hgihly risk patient

(D) Chronic SDH

1. Generally occurred in the elderly, other risk factories: alcohol abuse, seizure, CSF shunts, coagulopathy, patients at risk for falls
2. Usually associated with head trauma, even minor, weeks ago
3. Presentation : greatly uniform symptoms (minor symptoms as headache, confusion, language difficulty or coma, hemiplegia, seizure, focal sign ---)
4. Treatment options:
   (a) Burr holes drainage (two burr holes, or signle burr hole) with or without subdural drainage
   (b) Twist drill craniotomy
   (c) Formal craniotomy with excision of subdural membranes (hyperdensity or thickened wall lesions, multiple loculation, or unsatisfactory improvement after other treatment modality)

(E) Confessional ITCH

1. Coup lesion: contusions occurred at the site of impact
2. Contrecoup lesion: contusions distant from the site of impact (due to impacts of moving brain on the irregular surface of bony floor, eg. Frontal base & middle fossa)
3. Shearing force ICH: due to shearing force with rapid acceleration or deceleration of brain (multiple scattered ICH eg: in deep structure, corpus callostum, brain stem)
4. Immediate ICH
5. Delayed ICH – coalesce from “ salt and pepper” lesions to form large hematomas.
6. The decision to undertake surgical intervention is individualized, however not always easy. Generally, patient who are comatose and have a significant midline shift due to hematoma usually need surgery.

Special Consideration in Children Head Injury

(A) The outcome of pediatric moderately severe and severe head trauma is noted to be favorable in most modern reports.

(B) Growing skull fracture (post-traumatic leptomeningeal cysts)

1. Child ≤ 2 Y/O linear skull fracture → follow skull X-ray (AP + Lat) 8 weeks later
2. True post-traumatic leptomeningeal cysts need to be excised surgically with dural defect repair
3. However, " pseudo-growing fracture " should be treated conservatively

(C) Extra-axial fluid collections in children

1. Benign subdural collection in infants
2. Chronic symptomatic extra-axial fluid collections or effusion (hydroma)
3. Cerebral atrophy
4. External hydrocephalus
5. Acute subdural hematoma
6. Normal variant of enlarged subarachnoid spaces and inter-hemispheric fissure

(D) Child abuse

1. Usually had poor clinically outcome due to delayed treatment, seizure and / or hypoxic ischenic insult, and multiple prolonged injury
2. Factories which raise the suspicion
   (a) Retinal hemorrhage
   (b) Bilateral chronic SDH in child < 2 Y/O
   (c) Multiple skull fracture
   (d) Multiple or staged wounds within the body
3. Notify social worker

Spontaneous SAH

(A) Causes

1. Ruptured intracranial aneurysms: 75-80%
2. Cerebral AVM: 4-5%
3. Vasculitis in CNS
4. Tumor
5. Carotid artery dissection
6. Unknown

(B) Symptoms

1. Generalized headache with vomiting
2. Syncope
3. Cranial nerve palsy
4. Meningeal sign
5. Coma

(C) Grading Hunt & Heiss

Grade	
0	Unruptured aneurysm
1	Asymptomatic or mild H/A & mild unchal rigidity
1A	No acute meningeal/brain reaction but with fixed neuro deficit
2	Cranial N. palsy, moderate to severe H/S, nuchal rigidity
3	Mild focal deficit lethargy or infusion
4	Stupor, moderate to severe hemiparesis early decerebrate rigidity
5	Deep coma, decerebrate rigidity, moribund

* Add one grade for serious system disease (e.g. HTN, DM, COPD, Atherosclerosis) or severe vasospasm on angiography

(D) Admitting order

1. I/O, GCS, VS at least q.2.h. (in ICU-I or ICU-II)
2. Activity: Bed rest, avoid strong external stimulation (Loud noise, light visit)
3. Diet: NPO if lethargic or prepare to examination or OP. otherwise on diet as tolerable
4. IV: isotonic saline (correct volume & electrolyte)
5. Medication: (Avoid I.M. drug)
   (a) Slight sedation: Luminal 30-60 mg p.o./IV
   (b) Analgesia: codeine 30-60 mg p.o.p.r.n. or Demeral 25-50 mg i.v.
   (c) Stool softer
   (d) Anti-hypertensive drug
   (e) Anticonvulsant
   (f) Antiemetics
   (g) H2-blocker

(E) BP control (unclipped aneurysm)

	Day 0-3	Day 4-10	Day > 15
Systolic	120-140	140-160	120-140
diastolic	80-90	90-100	80-90

(F) Most likely complication

	Day 0-3	Day 4-14	Day > 15
Major complication	Brain edema Rebelled Acute hydrocephalus Cardic dysrhythmias Respiratory abnormalit	Cerebral vasospasm Rebelled Hypovolemia Hyponatremia Subacutehydrocephalus Pneumonia	Chronic hydrocephalus Pneumonia Rebelled Cerbral vasospasm Water & eledinlyte Disturbance

(G) Cerebral vasospasm

1. Maximally 6-8 days after SAH, resolved 12 days later or longer
2. Treatment
   (a) Early surgery
   (b) Pharmacological arterial dilator (Ca channel blocker, sympatholytics, papavrine)
   (c) Mechanical arterial dilatation
   (d) Volume expansion (+ induced arterial hypertension)
   (e) Remove blood clot
   (f) By-pass

Intubation and resuscitation in neurosurgical patient

呼吸道維護、輸液及監測系統

呼吸道臨床評估

由於呼吸道在外傷處置的重要性與急迫性，其評估幾乎全是就臨床表徵來判斷之，例如病人會呈現燥動不安或因血中二氧化碳上升而呈現意識遲鈍狀態;氣道中異常聲音或發生能力受阻；氣管的位置是否在正中線，胸壁上下起伏的程度，兩側肺野聲音是否聽的見，故臨床上評估呼吸道其技巧不外是眼道、耳到、手到，加上口到。如果没有反應表示病人神智改變，如果有而正確的聲音反應，表示其呼吸道通暢，通氣量足夠，腦部的血流灌流也足夠。

呼吸道維護與頸椎外傷

頸椎外傷一定要與呼吸維護一併考慮，臨床上才不致於顧此失彼。在企圖建立呼吸道時，不當的頸椎轉動而可能導致頸椎的移位或神經的進一步受損，如何建立氣道方式最可行。目前認為經口式氣管插管在執行時若能有第二人全程幫忙固定頸椎，且插管動作熟練，必要時加上短效神經肌肉阻斷劑的給予，應該可以將插管時頸椎受影響之程度減到最低。

呼吸道處置的適應性

呼吸道處置的適應症分成三類：〈1〉絕對必要：包含急性呼吸道阻塞、呼吸停止、缺氧、頸部穿刺傷併血腫塊，其中第四種情形最常發主在頸部遭受穿束傷時，血腫塊的擴大會引起氣道的阻塞,此一狀態若能及早予以氣管插應能在不是太困難時就解決該病人的問題。〈2〉相對必要：包含頸部外傷、休克、胸腔受損、大量後腹腔出血、瀕臨生命危殆的躁動外傷病人,在相對必要的情形可能是創傷間接的造成生理上或解剖上呼吸的損傷。〈3〉可能必要：包含顏面外傷、肺部挫傷等，在此情形下有可能初期不需要插管，但經過一段時間隨著病情的變化有可能需要插管。

輸液給予與監測系統

成人外傷病患在低血壓狀態時，首先靜脈注射２公升的林格氏液〈小兒科是以２０ml/kg 為輸液標準計量〉是一般公認的緊急輸液程序，之後視病患對此初始輸液的血壓反應，再決定追再與否。追加的輸液量在小兒科則是２０ml/kg 連續兩次，若是血壓仍然不穩定，則應即刻給予輸血或手術止血步驟，以便更積極挽救病患。依急救的進程而言，就過去２０年的各類文獻探討的結論，林格氏液以其能較有效維持並提升血壓特性而且取得容易價格便宜等優點，其重要性不容置疑，然而若只單純大量投予，則滲透壓的極巨改變，將導致血管內液體流向血管外，而滯積於組織間造成水腫，故依 ATLS 建議，應適當以３：１的輸液原則給予血漿或其代用品的 Colloid 溶液以平衡血管內外的滲透壓，使輸液能常留於血管中，有效維持血壓。但就外傷失血的病人而言，除了平衡滲透壓以外，血球比容與適當血紅素可攜帶氧分子到組織細胞的重要因素，因此輸血在外傷處置理扮演相當重要的角色。如今大家對創傷後輸血的觀念主要是以濃縮紅血球〈P-RBC〉為第一選擇。

頭部及脊椎創傷

頭部外傷

意外事故在台灣仍然是很重要的死因，頭部外傷雖然在安全帽法實施後大量減少，然而嚴重頭部外傷發生後之死亡率仍然高達 35%，如何使嚴重頭部外傷發生後得到最好的照顧實為當務之急。有關頭部外傷的治療理論最近二、三十年來有相當大的改變，由於對顱內壓升高的病理生理機制及腦部循環動力學的了解,相關的治療已有重大的突破。以往將治療的目標專注在處理顱內壓升高，例如: 限水(fluid restriction)、高張利尿劑(mannitol、glycerol)、過度通氣(hyperventilation)及類固醇等使用均被重新再評估。主要的改變是因腦部監測(cerebral monitoring)技術的進步以及預防腦部缺血(ischemia)為概念的發展。目前腦部重症加護照顧的中心目標已經由顱內壓的控制轉為缺血的預防，因此，除了降低顱內壓之外，如何提升腦部的灌流(cerebral perfusion)及降低腦部代謝以避免缺血，已成為另一注意的焦點。

輸液之原則 (Fluid Therapy)

原則

1. 作為大量輸液時 0.9% 生理食鹽水 (Saline) ， 較乳酸林格氏液 (Lactate Ringer's) 佳 (Level Ⅲ) 。
2. 含醣類輸液應避免使用 (Level Ⅲ) 。

建議(Options)：

1. Rosner et al 及 Lund concept 中皆提到使用膠質液 (colloid) 以維持血中之膨脹壓 (Level Ⅱ) 。
2. 有關膠質液 (colloid) 是否應該使用目前沒有足夠證據 。 但依國內專家學者共識嚴重頭部外傷患者認為膠質液 (Colloid) 可適當使用 。 至於使用那一類膠質液 (colloid) 仍未有共識 (Level Ⅳ) 。
3. FFP 祇適合凝血性病變 (Coagulopathy) ， 不建議使用在例行性容積擴張劑 (volume expander) (Level Ⅲ) 。

4. 頭部外傷併發休克的病患 ， 於到院前可使用高張鹽水 (Hypertonic saline) (Level Ⅱ) 。

插管藥物的選擇之使用

1. Atropine: 可以減少迷走神經對於插管的反應，同時也可減少唾液的分泌。
2. Lidocaine: 100mg IV push 可減少咽喉的反應。
3. Diazepam: 效果不定，可能會影發過長的 somnolence。
4. Midazolam: 水溶性，較穩定，可以有較好的鎮靜效果
5. Thiopental: 呼吸抑制，大劑量可造成呼吸停止，對於頭部外傷的病患或是顱內壓升高的病患可以此藥物使用
6. 止痛藥: 可以提供止痛效果及部分鎮靜效果,Morphine 和 fentanyl 這兩種成癮性止痛藥物除了有止痛，鎮靜的功能之外，這兩種藥物也有呼吸抑制以及止咳的作用，所以用於會對抗呼吸器的病人，有很大的幫助。因此，在輕中度頭部外傷合併多重創傷的病人，這兩種止痛藥的使用可以幫忙病人止痛，及不對抗呼吸器的效果，間接的也能達到避免顱內壓上升的目的。不過，其所造成腸胃道蠕動減緩的作用，所造成的便秘而引起的 Valsava 作用導致的顱內壓上升也是要謹慎注意的。此外，ketamine 的作用機制和其他類的加護病房用藥相當不同。Ketamine 是讓病人處於解離的狀態而使病人有鎮靜止痛的效果。在傳統上，ketamine 被認為會增加顱內壓，血壓，腦血流。因此在神經重症上很少被利用到。
7. 肌肉神經阻斷劑: 在神經重症常用的神經肌肉阻斷劑包括去極化和非去極化阻斷劑兩大類。其中 succinylcholine 為去極化阻斷劑，一分鐘開始作用、維持五至十分鐘，主要應用在氣管插管(intubation)。

Brain Tumor

CLASSIFICATION

Tumor of neuroepithelial tissue: intrinsic, intra-axial, including: astrocytoma, oligodendroglioma, ependymoma, choroid plexus tumor, neuronal tumors, pinealocytoma/pinealoblastoma, embryonal tumors, embryonal tumors

Tumor of meninges: meningioma, hemangiopericytoma, melanoma

Tumor of cranial and spinal nerves: neurofibroma, neurinoma, MPNST

Lymphoma, Germ cell tumors, Cystic tumors, Tumors of the sellar region

PRESENTATIONS

Headache: 可能因為 IICP 造成, 可能早上比較厲害(hypoventilation during sleep), cough, valsava maneuver 可能會加劇症狀, 和 tension headache/migraine 需要做鑑別診斷, focal neurologic deficits and seizures 通常在 brain tumor 比較常看到

Seizures: 較少成為剛開始發現之主訴, 但一旦在病房遇到, 特別是第一次 onset 的病人需要積極處理, 第一線用藥目前以短效的 BZD 為主 (lorazepam iv 為首選也是某些病房常備藥), 請使用 iv push 給藥, 注意呼吸

Focal neurologic deficits: most common. 需要 conirm 跟 lesion 位置是否一致, 是否為 new onset 之症狀,

Mental status changes: 必須和其他 medical condition 造成的 conscious change 做鑑別診斷,

下面這兩種主要在 pituitary tumor 看到

Endocrine disturbances

Pituitary apoplexy: “hemorrhagic stroke” of pituitary tumor, 可能會合併視野改變, conscious change, sudden onset of headache, acute endocrine deterioration, 必須積極處理, 包括 glucocorticoids 使用及 surgical decompression

PERIOPERATIVE ORDERS

1.Shaving or not? 傳統的 craniotomy 都必須剃頭, 如果某些小傷口的手術(特別位 在 posterior fossa), 如 microvascular decompression, 可以考慮不剃頭到開刀房剃, 通常這種 case 主治醫師都會特別註明

2.Steroids in brain tumor: 對於 brain tumor 造成的 vasogenic edema 十分有效, 在急性期是降 ICP 的好方法, 主要有效的是 glucocorticoid effect, 所以臨床上我們常用的是 dexamethasone, 正常大人常用的劑量是 5-10mg iv loading and 5mg q6h. preop 的 order 通常會希望能帶 1-2 劑進開刀房.

此外,在 pituitary tumor 這種可能會影響正常生理 cortisol 分泌的手術, 術前 6-12 小時可以給病人用 solucortef 作 stress condition 底下的 supplement

3.Prophylactic anticonvulsants in brain tumor: 如果病患 initial 就是以 seizure 作為 表現, 請將原本在使用的 antiepileptics 帶 1-2 劑進到開刀房 (常見 iv 使用的是 dilantin 跟 deparkine), 如果病患原本沒有在使用 antiepileptics, 腫瘤開顱手術的預防性 antiepileptics 請帶 dilantin(phenytoin) 250mg x 3pcs (17mg/kg for loading) 通常足夠

4.Other perioperative orders in CGMH OR

還有一些藥物可能會使用到需要帶進刀房, Surgicel (人工合成的 cellulose, for hemostasis, 通常會用到), 2% lidocaine for subcutaneous injection(非必需)

依照常見住院接受腫瘤手術分類大致分成四類

1.Craniotomy:

剃頭; 1st generation cephalosporin; Dexan; mannitol (有時術中對 brain 的 manipulation 會讓 brain 顯得腫脹); Dilantin; Surgicel; lidocaine; blood preparation (FFP pRBC)

2.Stereotaxic Biopsy

視 lesion 位置, 大部分不需要剃頭 (術中剃), 1st generation cephalosporin, dexan, mannitol 通常不需要;不需要備血

3.Transphenoid approach

不需剃頭; Preoperative solucortef (200mg in iv 500cc run 20 cc/hr; 前一天晚上 12 點開始使用,通常會使用到術後 24 hrs) ; 1st generation cephalosporin; 通常不需要備血

4.Microvascular decompression or other minimal invasive craniotomy

不需剃頭; 1st generation cephalosporin; Dexan; mannitol (有時術中對 brain 的 manipulation 會讓 brain 顯得腫脹); Surgicel; lidocaine; 通常不需備血

IN-HOSPITAL CHEMOTHERAPY FOR GLIOMA

Temozolomide (Temodal) : FDA approved oral akylating agent after glioma recurrence. RX: 150 mg/m2/d po qd x 5 days per month. **此藥物乃專案申請藥物,晚上接到需要住院使用這種藥物的病人請不要自己開藥,請交給當科醫師或 NSP 開立(此藥物電腦一點出來就視同使用過,請連點都不要點)

Avastin: monoclonal antibody for VEGF, off-label use in glioma for advanced edema, 只有極少數病人會使用,須化療藥囑,請待當科照會腫瘤科